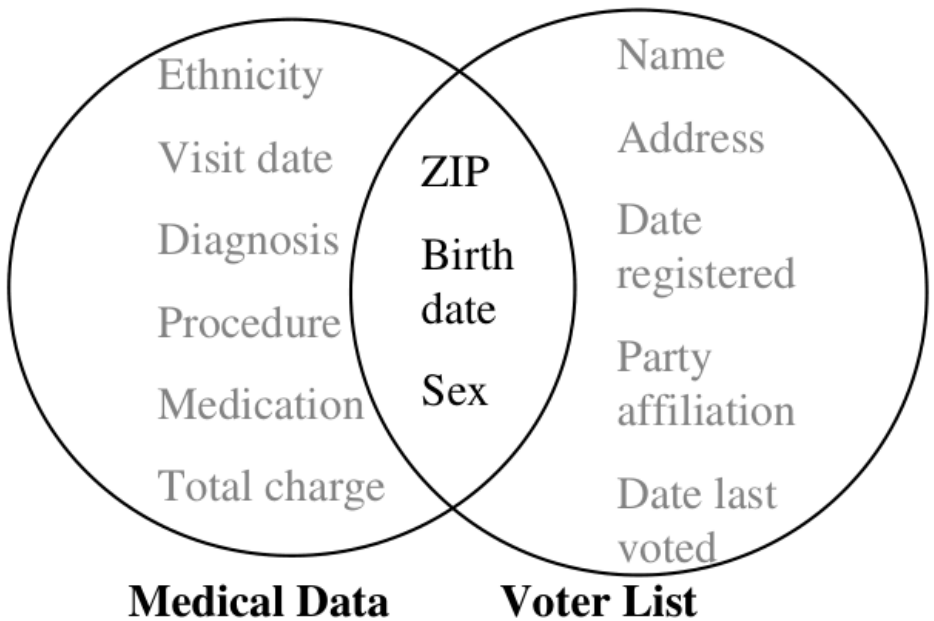
# 5 Data memorization and private learning

# Data release and personal privacy

**1990s:** growing digital records make useful analysis—and record linkage—easier.




Removing names leaves shared attributes that can identify people.

Sweeney (2002), Example 1

# Quasi-identifiers

**Synthetic example:** a medical release has no names.

| Age | Postal prefix | Diagnosis |
|---|---|---|
| 31 | N2L | Asthma |
| 37 | N2L | Migraine |
| 54 | N2J | Diabetes |
| 58 | N2J | Asthma |

A public profile says **Alex is 31 and lives in N2L**.

Age and location jointly link Alex to a diagnosis.

# $k$-anonymity

Every released quasi-identifier pattern occurs in **at least $k$ records**.

| Age | Postal prefix | Diagnosis |
|---|---|---|
| 30–39 | N2L | Asthma |
| 30–39 | N2L | Migraine |
| 50–59 | N2J | Diabetes |
| 50–59 | N2J | Asthma |

$k = 2$, for the specified quasi-identifiers: age band and postal prefix.

Sweeney (2002), Definition 3



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# Attribute disclosure

**Synthetic release:** two people share the same quasi-identifiers.

| Age | Postal prefix | Diagnosis |
|---|---|---|
| 30–39 | N2L | Asthma |
| 30–39 | N2L | Asthma |

> What can we infer about Alex if we know Alex is in this group? 🤔
>
> Answer in class.

The diagnosis is revealed even though the matching row is ambiguous.

# Netflix Prize

**Released data:** user IDs, movies, ratings, dates.

**Auxiliary information:** some ratings associated with a named person.

$$\underbrace{\text{a few known ratings}}_{\text{outside the release}} \longrightarrow \underbrace{\text{a longer viewing profile}}_{\text{inside the release}}$$

Sparse, high-dimensional records can be distinctive.

Narayanan and Shmatikov (2008), §5

# Data, models, and guarantees

$$\text{Private records} \xrightarrow{\text{release}} \text{Dataset}$$

Linkage and attribute disclosure can happen here.

$$\text{Private records} \xrightarrow{\text{training}} \text{Model} \xrightarrow{\text{query}} \text{Output}$$

Training can retain details that an output reveals.

**Differential privacy** constrains the randomized release procedure—whether its output is a statistic, dataset, or model.

# Training-data memorization

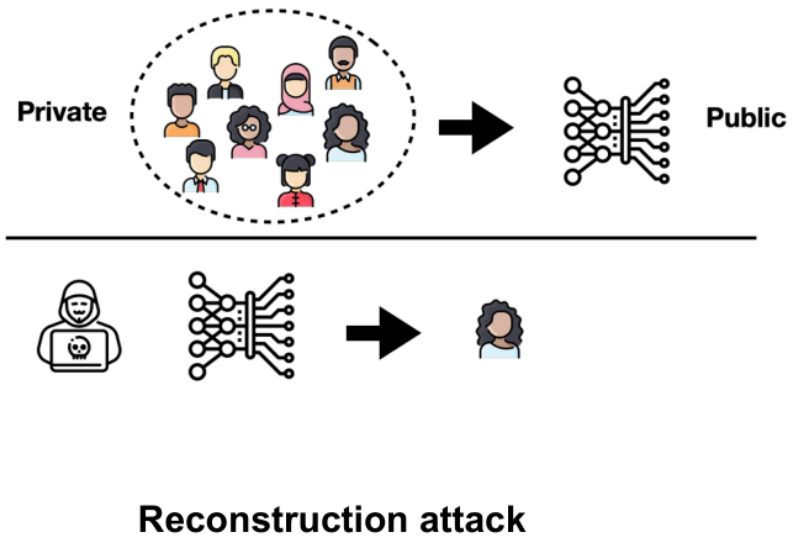
## Illustrative private training text

> My recovery phrase is amber fern cobalt meadow.

**Prompt:** My recovery phrase is …

**Possible completion:** amber fern cobalt meadow.

Exact reproduction can expose a training example.




Carlini et al. (2021), §§3–5

# Overparameterized linear regression

$$X \in \mathbb{R}^{n \times d}, \quad y \in \mathbb{R}^n, \quad \hat{w} \in \arg\min_w \|Xw - y\|_2^2.$$

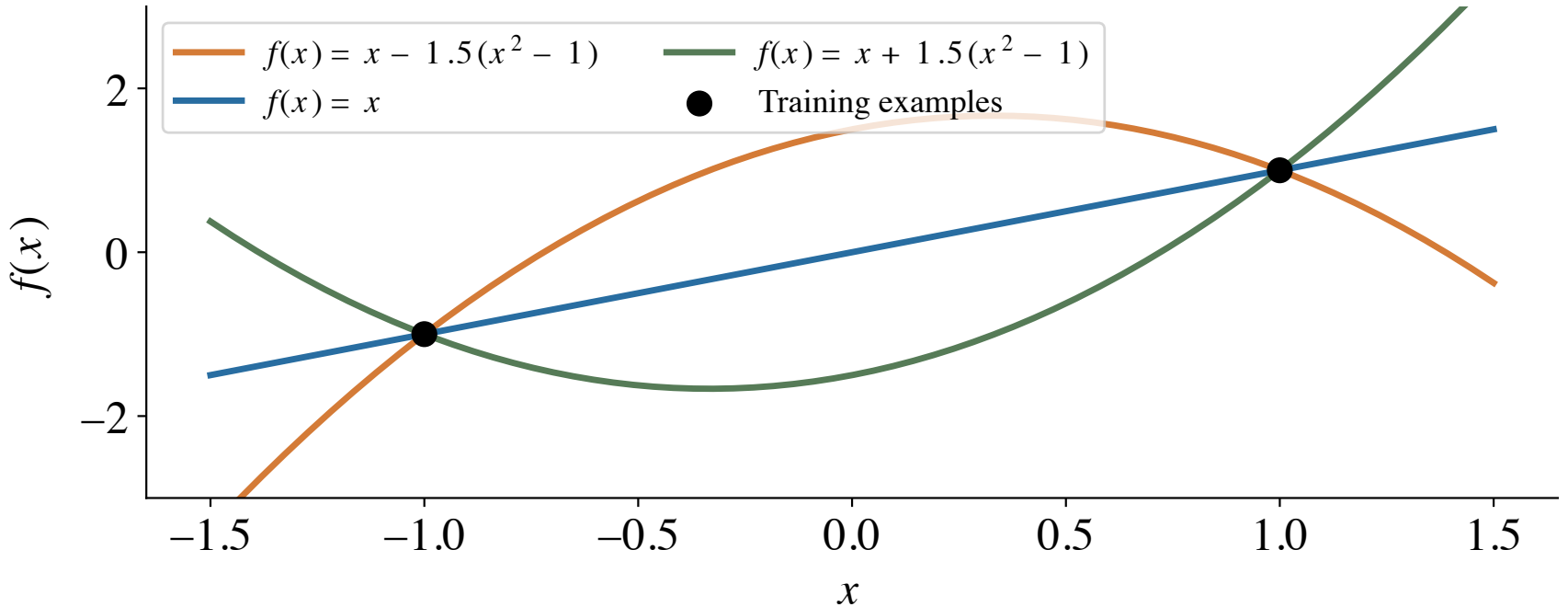
For $d > n$ and $\mathbf{rank}(X) = n$, every $y$ can be fitted exactly.

$$\underbrace{w_0 = X^\top (XX^\top)^{-1} y}_{\text{one solution}}, \qquad \underbrace{w = w_0 + v, \quad Xv = 0}_{\text{all interpolating solutions}}.$$

The training data leave $d - n$ **unconstrained directions**.

# Interpolation and generalization

Two training examples; three coefficients: $f(x) = w_0 + w_1 x + w_2 x^2$.

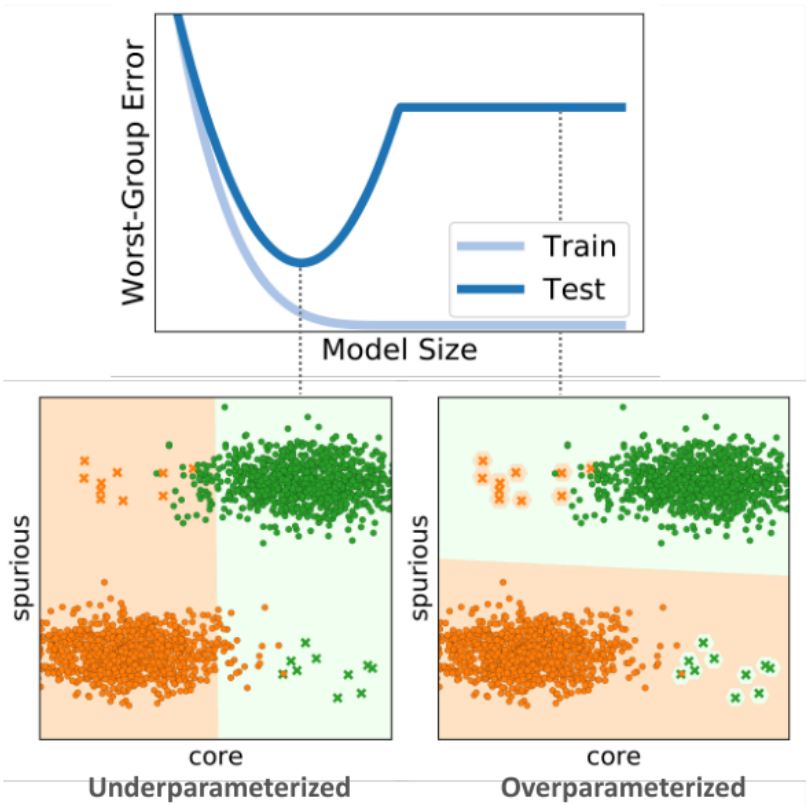



All three predictors have **zero training error**.



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# Spurious correlations and memorization




Sagawa et al. (2020), Fig. 1

**Core feature:** useful across groups.

**Spurious feature:** correlated with the label in the majority.

Extra capacity can fit the minority training examples while relying on the spurious feature.

# The privacy unit

$$\text{one message} \subset \text{one conversation} \subset \text{one person's history}$$

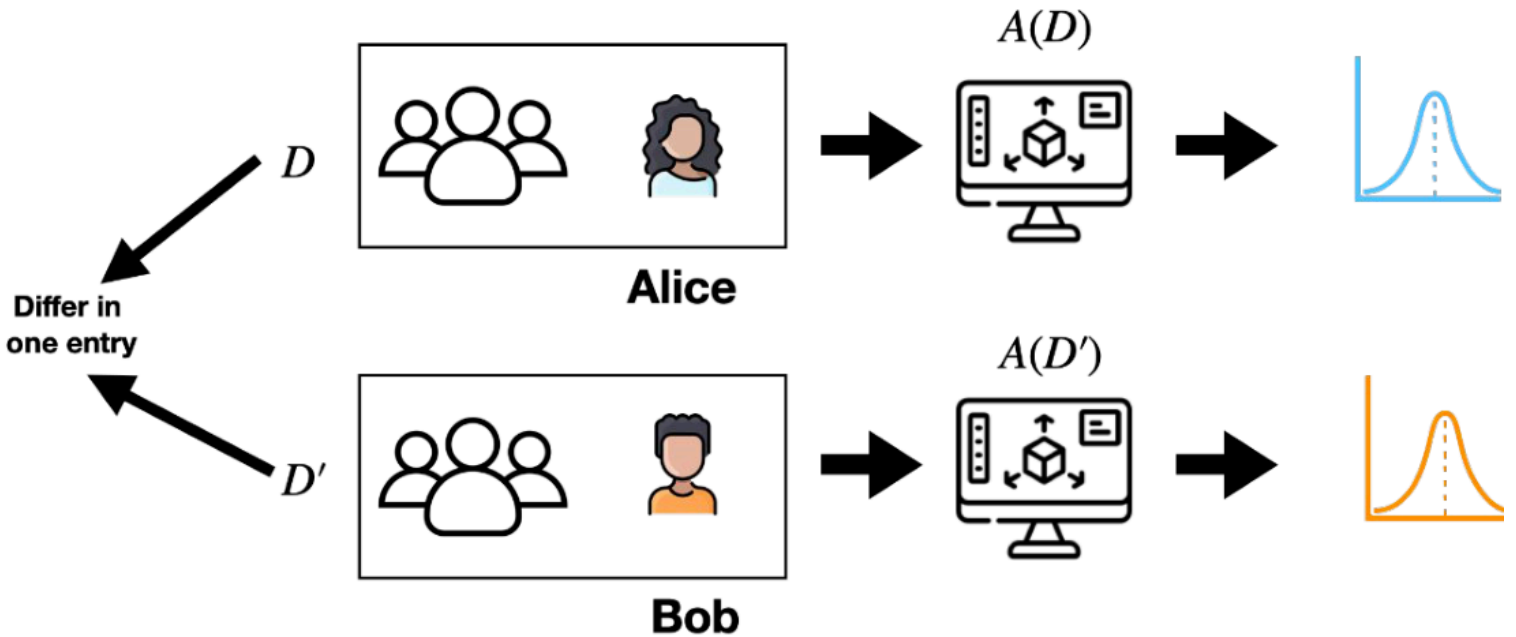
**Neighboring datasets** differ by the contribution we want to protect.

For this lecture: $D \sim D'$ means **add or remove one person's contribution**.

One row per person in the counting example.

Dwork and Roth (2014), §2.3

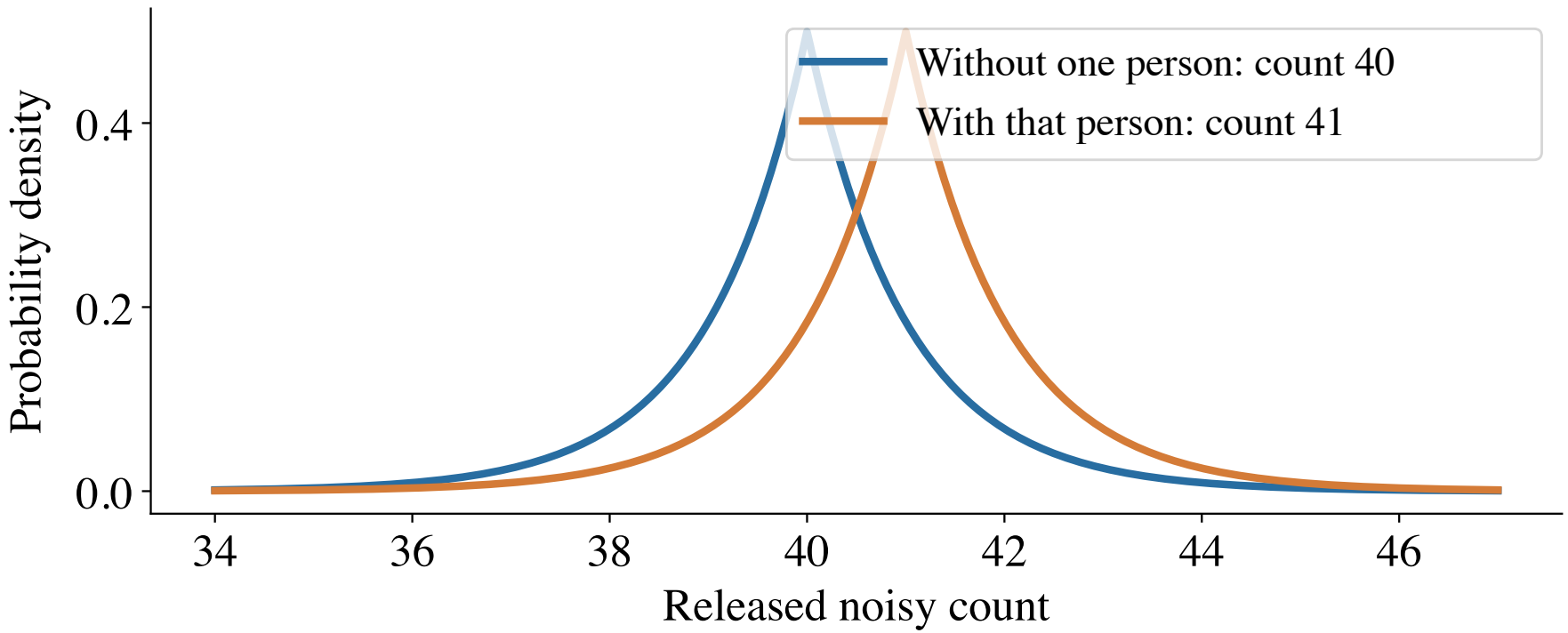
# Neighboring datasets




**Replace one record:** Alice's record is replaced with Bob's.

**Add/remove adjacency:** compare a dataset with and without one contribution.

# Randomized releases




Compare **distributions over outputs**, with the input dataset held fixed.

# Pure differential privacy

A randomized mechanism $M$ is **$\epsilon$-differentially private** if

$$\Pr[M(D) \in S] \;\leq\; e^{\epsilon} \Pr[M(D') \in S]$$

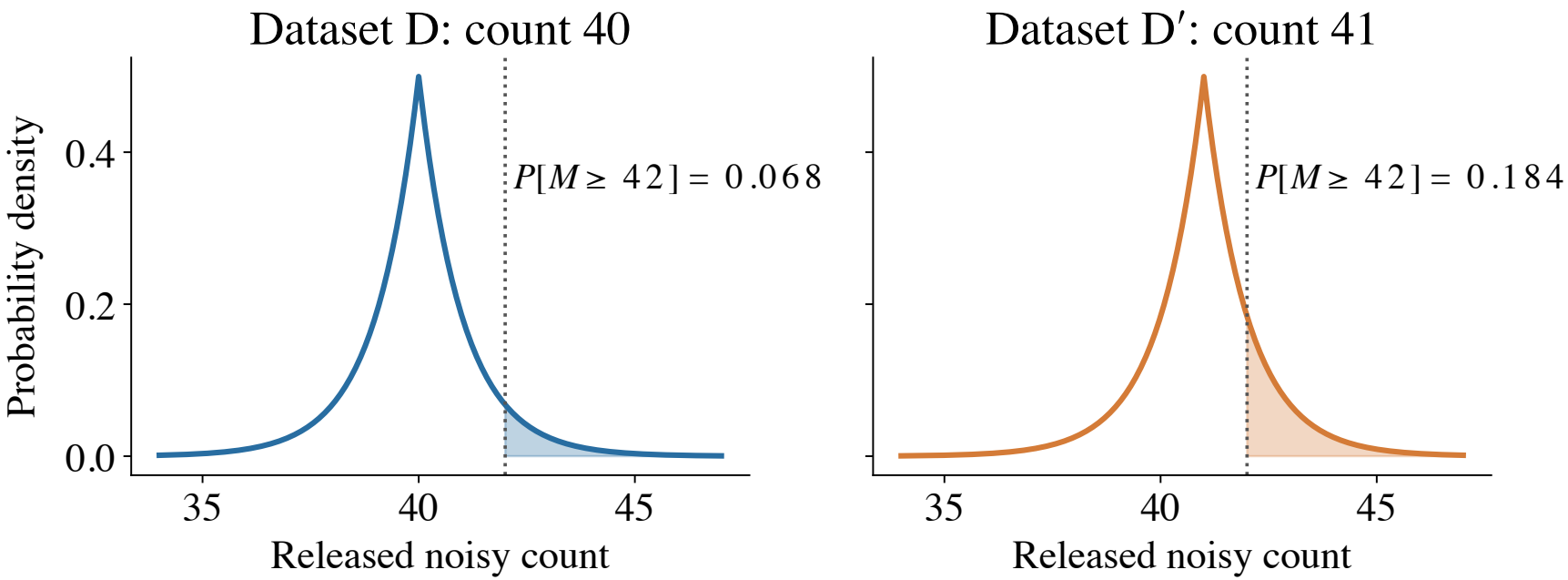
for **every** neighboring pair $D \sim D'$ and **every** measurable output event $S$.

- Probabilities are over the mechanism's randomness.
- Swapping $D$ and $D'$ gives the reverse bound.
- Smaller $\epsilon$ requires more similar output distributions.

Dwork and Roth (2014), Definition 2.4 with $\delta = 0$

# Output events

Event $S$: **the released count is at least 42.**




$$\frac{\Pr[M(D') \geq 42]}{\Pr[M(D) \geq 42]} = e \approx 2.72.$$



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# Approximate differential privacy

$$\Pr[M(D) \in S] \leq e^{\epsilon} \Pr[M(D') \in S] + \delta.$$

$(\epsilon, \delta)$**-DP** allows additive slack in the event-probability bound.

$\delta = 0$ recovers pure DP.

Small $\delta > 0$ permits mechanisms such as appropriately calibrated Gaussian noise.

Both parameters belong in the reported guarantee.

Dwork and Roth (2014), Definition 2.4

# Sensitivity

For a scalar query $f$, its **global sensitivity** is

$$\Delta f = \sup_{D \sim D'} |f(D) - f(D')|.$$

**Count:** $f(D) = \sum_{i \in D} \mathbf{1}\{\text{person } i \text{ used the service}\}$

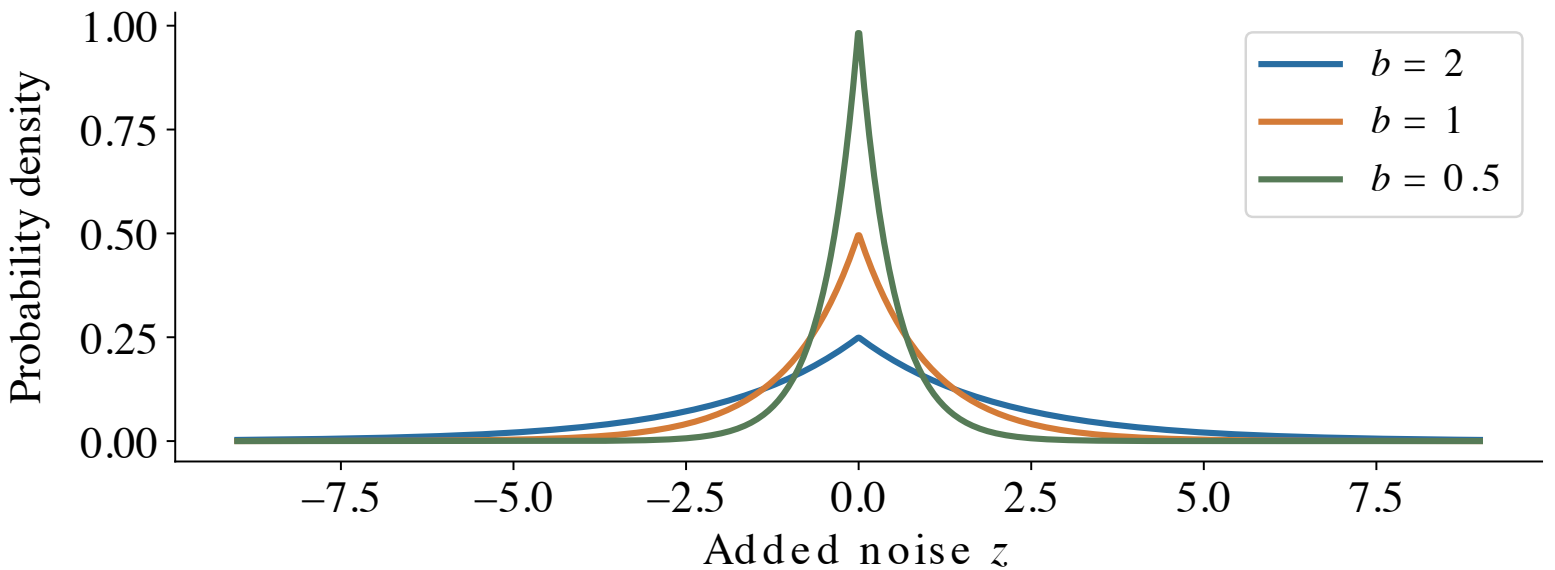
$$\Delta f = 1.$$

**Total messages:** unbounded unless each person's contribution is bounded.

Dwork and Roth (2014), Definition 3.1

# The Laplace mechanism

$$M(D) = f(D) + Z, \qquad Z \sim \mathrm{Lap}(0, b), \qquad b = \frac{\Delta f}{\epsilon}.$$




$$p_Z(z) = \frac{1}{2b} \exp\left(-\frac{|z|}{b}\right).$$

Dwork and Roth (2014), §3.3

# Laplace mechanism: proof

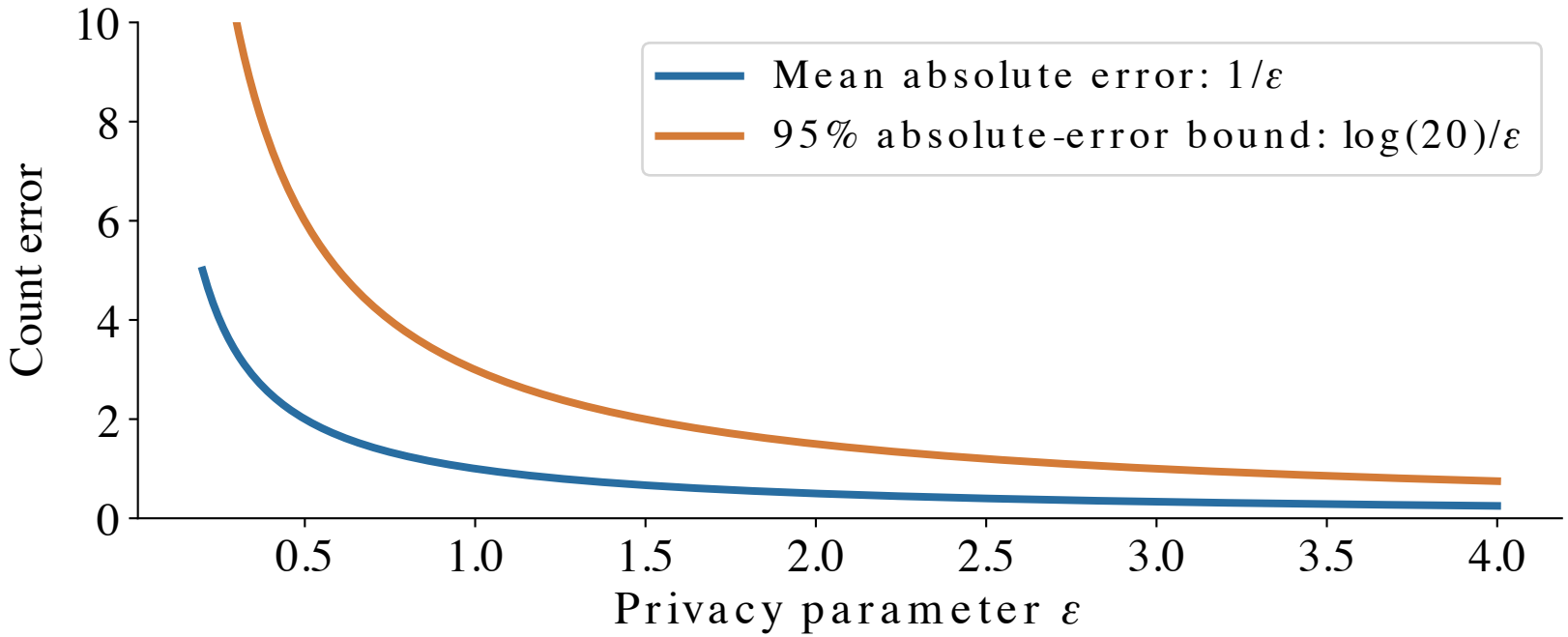
Write $p_D(y)$ for the density at released value $y$, and $b = \Delta f / \epsilon$.

$$\begin{aligned}\frac{p_D(y)}{p_{D'}(y)} &= \exp\left(\frac{|y - f(D')| - |y - f(D)|}{b}\right) \\ &\leq \exp\left(\frac{|f(D) - f(D')|}{b}\right) \\ &\leq \exp(\Delta f / b) = e^{\epsilon}.\end{aligned}$$

Integrate over any event $S$: $\Pr[M(D) \in S] \leq e^{\epsilon} \Pr[M(D') \in S]$.

Dwork and Roth (2014), Theorem 3.6

# Privacy and count accuracy




$$\mathbb{E}|Z| = \frac{1}{\epsilon}, \qquad \Pr\left(|Z| \leq \frac{\log 20}{\epsilon}\right) = 0.95.$$

One count, one contribution per person, $\Delta f = 1$.

# Auxiliary information

Hold the other records and the attacker's information fixed.

$$\frac{\text{posterior odds of } D \text{ versus } D'}{\text{prior odds of } D \text{ versus } D'} = \frac{p_D(y)}{p_{D'}(y)} \leq e^{\epsilon}.$$

The release adds bounded evidence between these neighboring possibilities.

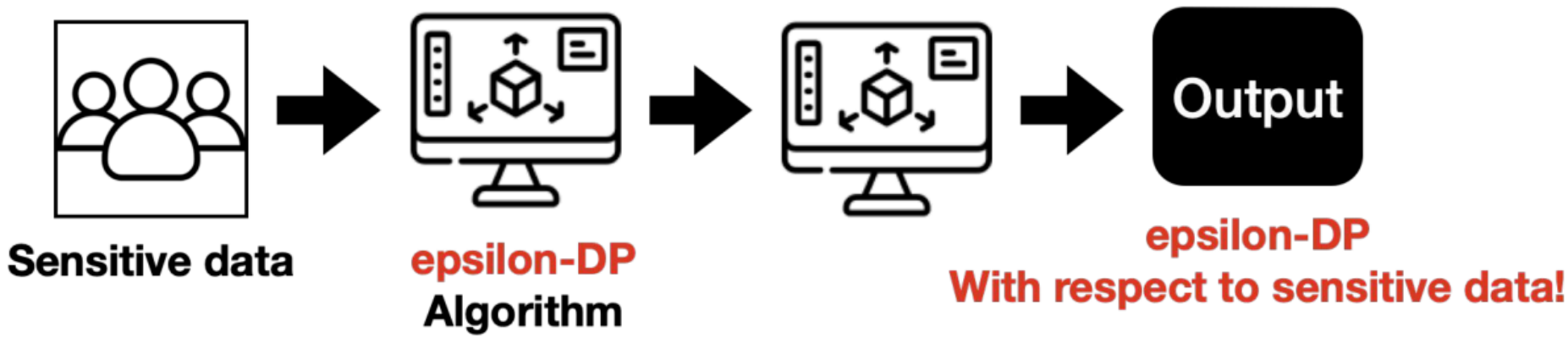
Background knowledge can still reveal facts about a person.

Dwork and Roth (2014), §§2.3, 2.3.1



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# Post-processing



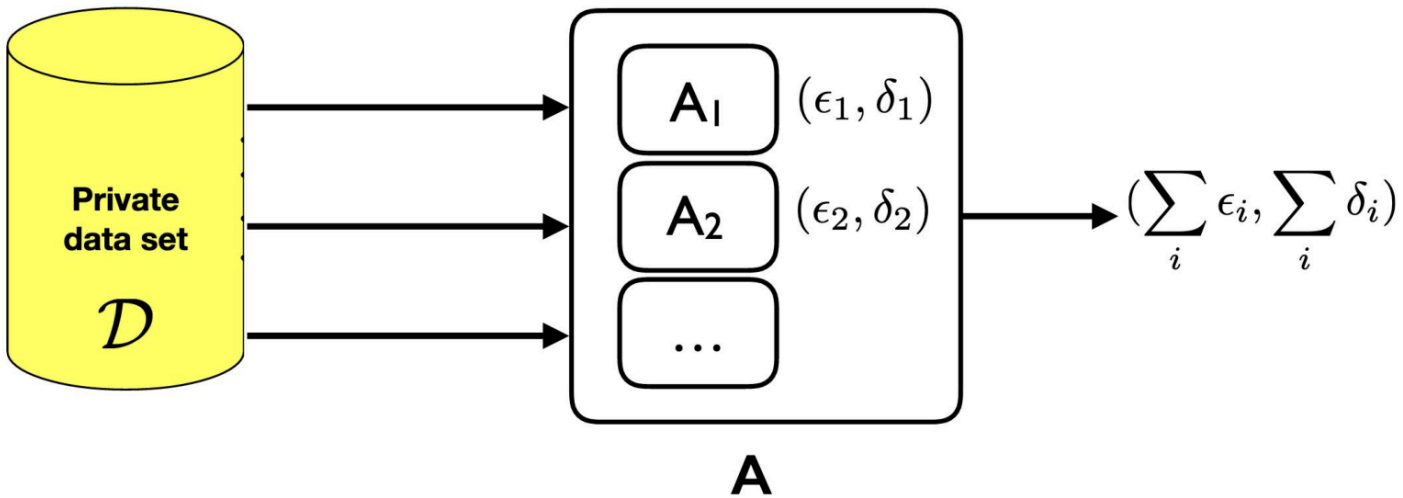

If $M$ is $(\epsilon, \delta)$-DP, then $g \circ M$ **has the same guarantee**.

Round a noisy count. Draw a chart. Query a released DP model.

$g$ must not access additional private data.

Dwork and Roth (2014), Proposition 2.1

# Composition




If each conditional step is $(\epsilon_t, \delta_t)$-DP, the full transcript is

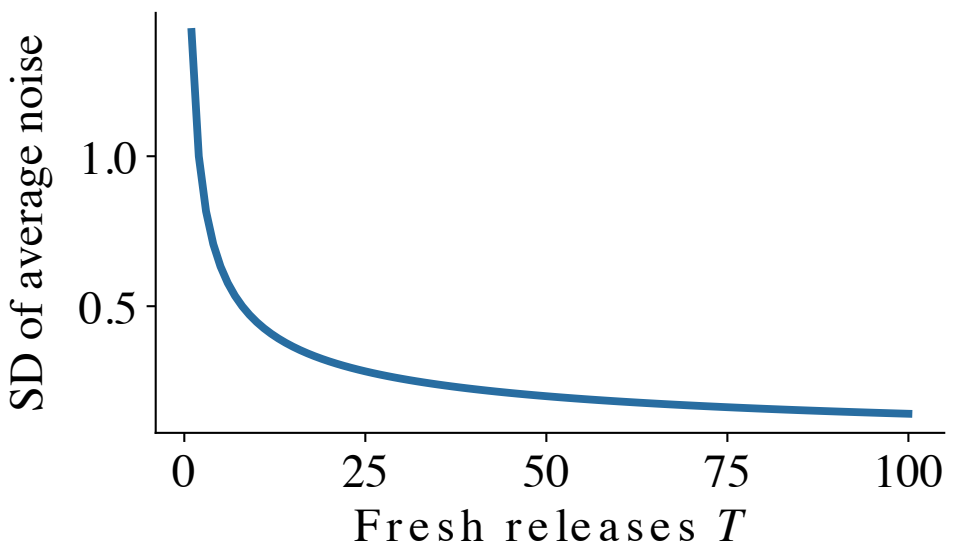
$$\left(\sum_t \epsilon_t, \sum_t \delta_t\right) \text{-DP}.$$

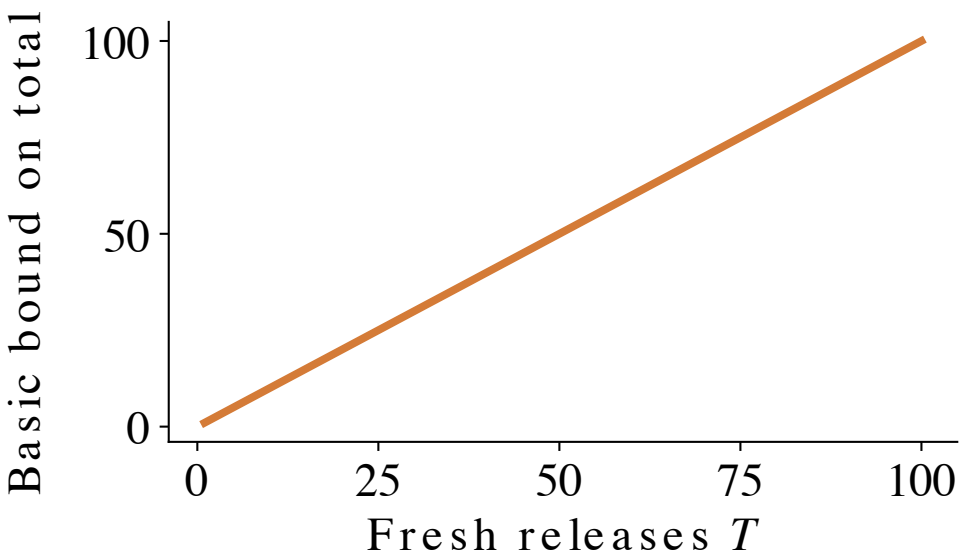
Build complex mechanisms from private components—and account for the whole transcript.

Dwork and Roth (2014), §3.5 and Appendix B

# Repeated noisy counts

Release $f(D) + Z_t$ repeatedly with independent $Z_t \sim \mathrm{Lap}(0, 1)$.



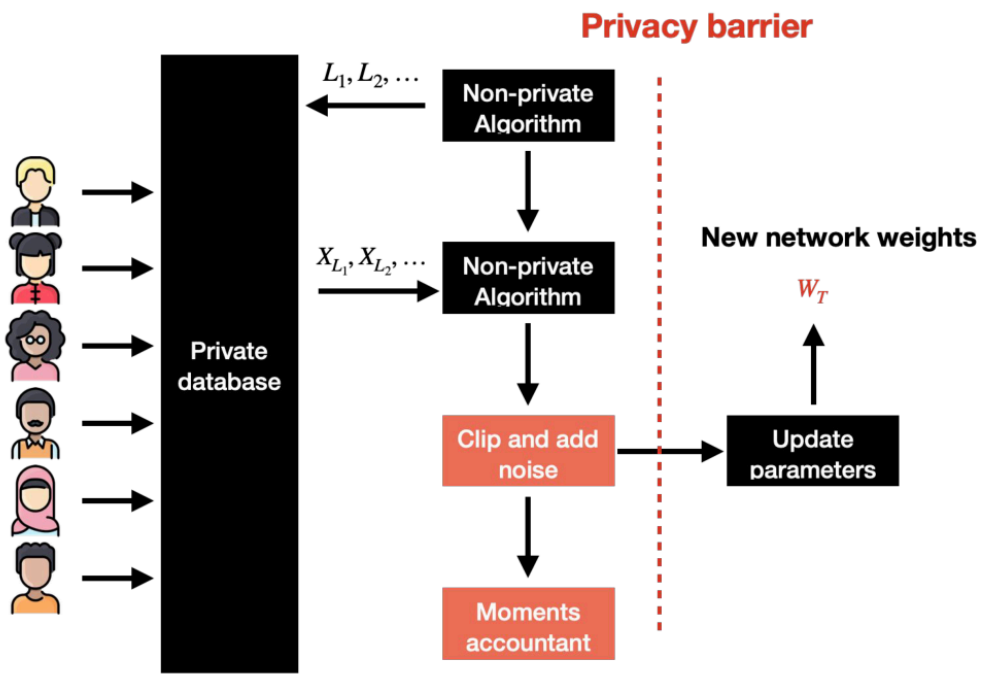




$$\mathrm{SD}\left(\frac{1}{T}\sum_t Z_t\right) = \sqrt{\frac{2}{T}}, \qquad \epsilon_{\text{total}} \le T.$$



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# Private model training




Abadi et al. (2016), §3

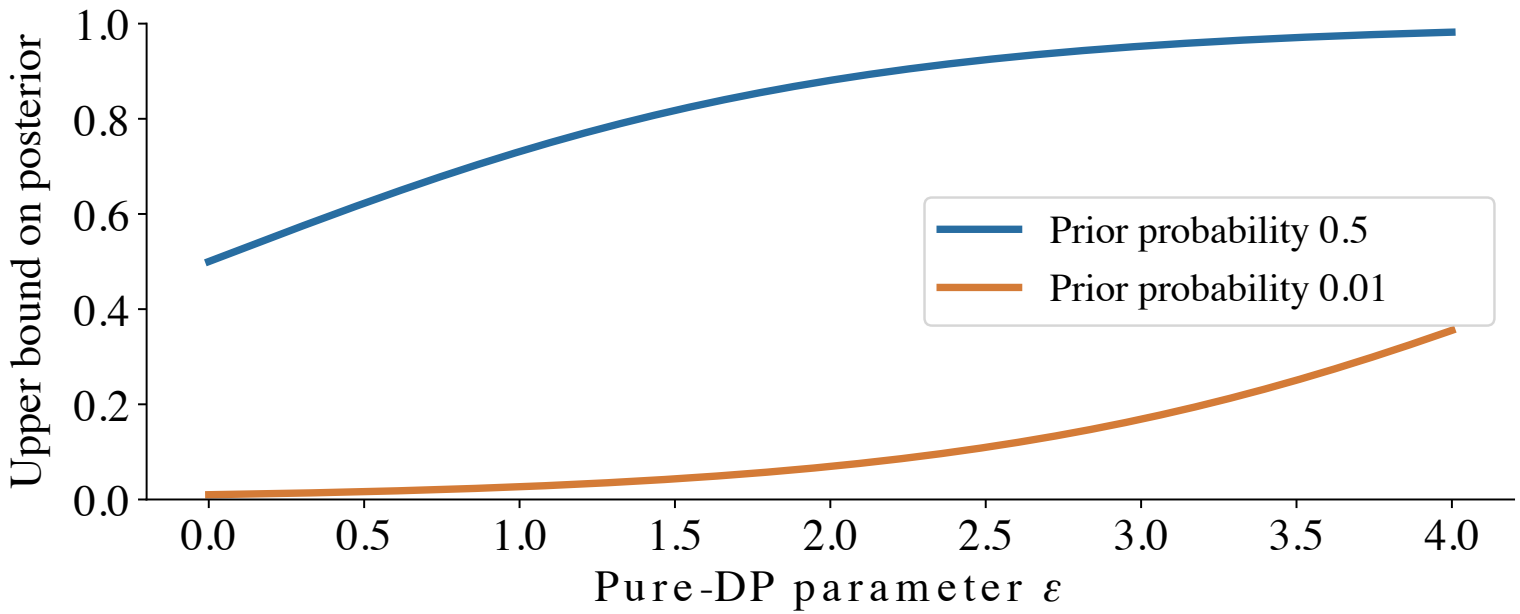
**Sensitivity:** bound one example's contribution.

**Composition:** account across training steps.

**Release:** parameters and any other disclosed outputs.

# Interpreting $\epsilon$

For pure DP, $e^{\epsilon}$ bounds the change in odds between fixed neighbors.




$\epsilon = 1$: odds multiplier at most **2.72**. $\epsilon = 2$: at most **7.39**.



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# A homework chatbot

**Hypothetical proposal:** train on student conversations and release the model with $(\epsilon = 2, \delta = 10^{-6})$-DP.

> Would you accept this use of your conversations? What else would you need to know? 🤔
>
> Answer in class.

Privacy unit · repeated training · access to raw text · usefulness · alternatives

# The 2020 U.S. Census

**Redistricting statistics:** useful small-area counts with formal disclosure protection.

$$\text{Confidential records} \longrightarrow \text{noisy measurements} \longrightarrow \text{consistent tables}$$

The June 2021 announcement reported a total budget of $\epsilon = 19.61$ for the redistricting product.

Interpreting it also requires the **privacy definition, $\delta$, protected unit, and invariants**.

U.S. Census Bureau, June 9, 2021 · TopDown algorithm

# Differential privacy in industry

**Apple:** local DP for aggregate usage statistics; its published overview specifies per-donation budgets and contribution limits.

**Microsoft:** a differentially private U.S. broadband-coverage data release.

> What does "uses differential privacy" leave unspecified? 🤔
>
> Answer in class.

Apple, technical overview · Microsoft Research, 2021

# Central and local privacy

## Central DP

$$\text{People} \xrightarrow{\text{raw data}} \text{trusted curator} \xrightarrow{\text{DP mechanism}} \text{release}$$

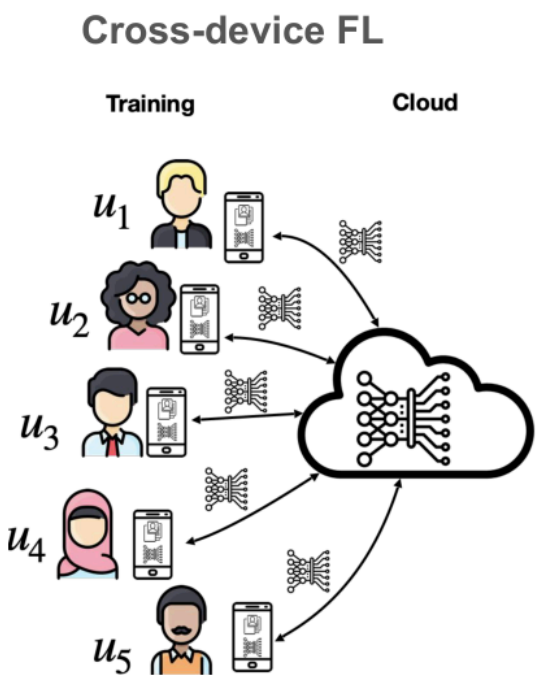
## Local DP

$$\text{Person's data} \xrightarrow{\text{local randomizer}} \text{message} \longrightarrow \text{collector}$$

The observer—and the point where protection begins—changes.

Dwork and Roth (2014), Chapter 12

# Federated learning




Training data stay on devices; **model updates are communicated**.

Federated learning alone provides **no DP guarantee**.

User-level DP additionally needs bounded user influence, calibrated noise, and accounting across rounds.

B. McMahan et al. (2017); H. B. McMahan et al. (2018)

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